Marker Length Study: Stage 4b Results

FYP, DNA data storage · 28 October 2026

Longer markers recover more strands, but most lengths only break even on stored data; a **6-base marker** gave the best balance, recovering **3.9% more data** than no markers at 8% deletion with 25% less overhead than the 8-base marker.

Question and setup

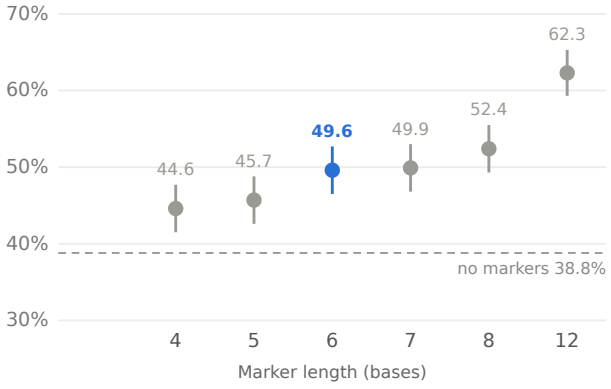
Can a shorter marker keep the benefit of the 8-base marker while using less space, and does a longer one add anything?

- **Layout:** every strand is 128 bases with 4 markers, one at the end of each 32-base block. A marker of L bases leaves 32 − L data bases per block, so a shorter marker means more data per strand.
- **Same everything else:** same random payloads, the same error positions in every read, 10 reads per strand, deletion 1/3/5/8% with 1% insertions and substitutions.
- **Same training:** each version fine-tuned from the same pretrained DNAFormer with the locked Stage 4 settings (10 epochs, seed 1), best epoch by validation, scored once on 1,000 test strands per level.
- **Winner rule (agreed before the results):** most data recovered at 8% deletion; ties go to the shorter marker.

Results

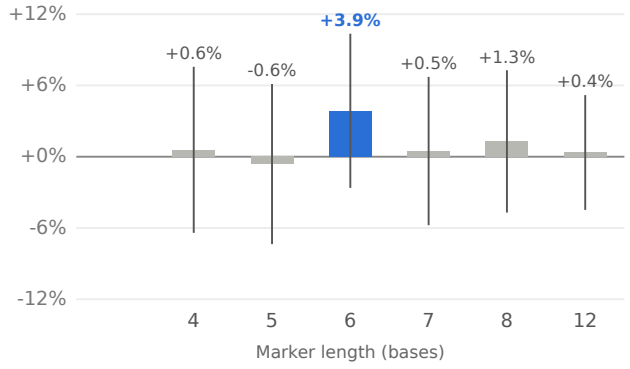
Longer markers recover more strands

Exact strand recovery at 8% deletion, with 95% CI



Only length 6 clearly stores more data

Data recovered at 8% vs no markers (exact strands x data bases)



Left: blue = length 6; dashed line = no markers. Right: bars = change in data recovered versus no markers; whiskers come from the 95% CI of each version's exact recovery.

Results table

Version	Marker	Data bases	Overhead	Exact 5%	Exact 8% [95% CI]	NED 8%	Break-even at 8%	Data recovered 8% (bases per 1,000 strands)	vs no markers
No markers	-	128	0.0%	94.1%	38.8% [35.8, 41.9]	0.0202	–	49,664	–
Length 4	CAGT	112	12.5%	95.4%	44.6% [41.5, 47.7]	0.0183	44.3%	49,952	+0.6%
Length 5	TCATG	108	15.6%	95.6%	45.7% [42.6, 48.8]	0.0182	46.0%	49,356	-0.6%
Length 6	CTGTCA	104	18.8%	95.3%	**49.6%** [46.5, 52.7]	0.0172	47.8%	**51,584**	**+3.9%**
Length 7	CTGATCA	100	21.9%	96.1%	49.9% [46.8, 53.0]	0.0168	49.7%	49,900	+0.5%
Length 8	ACGTCAGT	96	25.0%	96.1%	52.4% [49.3, 55.5]	0.0148	51.7%	50,304	+1.3%
Length 12	ACGTCAGTGCAT	80	37.5%	97.1%	62.3% [59.3, 65.3]	0.0119	62.1%	49,840	+0.4%
End-block 8 (control)	*ACGTCAGT*	*96*	*25.0%*	*95.7%*	*41.0% [38.0, 44.1]*	*0.0241*	*51.7%*	*39,360*	*-20.7%*

Fine-tuned DNAFormer, 1,000 unseen test strands per deletion level, markers removed before scoring. Break-even = exact recovery needed to store as much data as no markers. Data recovered = exact strands × data bases.

Findings

1. **Accuracy rises with marker length.** At 8% deletion exact recovery goes 44.6% (4), 45.7% (5), 49.6% (6), 49.9% (7), 52.4% (8), 62.3% (12). Lengths 4–5 and 6–7 behave alike; the clearest step is between 5 and 6.
2. **On stored data almost every length breaks even.** The accuracy gained roughly equals the space given up (−0.6% to +1.3%). **Length 6 is the exception at +3.9%** and wins under the agreed rule.
3. **Markers pay off only under heavy deletion.** At 5% deletion every version recovers 94–97% of strands, so no markers stores the most data there and each marker version loses roughly its overhead.
4. **Placement still matters.** The end-block control with the same 8-base markers (41.0%) stays far below periodic markers (52.4%), so the gain comes from synchronization, not from the shorter payload.
5. **Training was healthy for all versions:** validation improved steadily and test results matched validation within normal variation.

Caution: one training run per version. Length 6's lead in stored data (+3.9%) is within the run-to-run uncertainty (whisker in the right chart); Stage 5 adds paired tests on the same strands and repeat runs to confirm it.

How the markers were chosen

Marker length decides how often a marker appears by accident in the data; within one length, the exact sequence was chosen by fixed design rules and checked against the dataset.

Markers used

Length	Marker	GC content	Strands containing it by chance*	Rank among eligible markers (rarest = 1)	Overhead	Data bases	Role
4	CAGT	50%	38.6%	1 of 40	12.5%	112	shortest tested
5	TCATG	40%	10.8%	1 of 184	15.6%	108	
6	**CTGTCA**	**50%**	**2.6%**	**2 of 304**	**18.75%**	**104**	**best balance**
7	CTGATCA	43%	0.5%	1 of 1,440	21.9%	100	
8	ACGTCAGT	50%	0.2%	–	25%	96	original design (Stage 4)
12	ACGTCAGTGCAT	50%	0%	–	37.5%	80	upper-bound check

*Share of the 15,000 marker-free 128-base payloads of the dataset that contain the marker anywhere. Full counts for every possible marker of length 2–7 are in marker_occurrence_table.xlsx.

Selection rules

1. **No repeated letter** (such as AA): nanopore sequencing often deletes letters inside repeats.
2. **No self-overlap** (such as ACAC): a marker that matches itself when shifted can be found in the wrong place.
3. **GC content 40–60%**: balanced markers are easier to synthesize and sequence.
4. **Lowest occurrence** in the dataset among the markers that pass rules 1–3. Re-tested on two new random datasets, this ranking did not repeat, because the data is random: within one length all eligible markers are about equally rare. Length is what matters: each extra letter makes accidental matches about 4× rarer.

Why not lengths 2 and 3

- **Length 2** (16 possible markers): every one of them appears in 99.96% of strands, about 8 times per strand, so it cannot mark a unique position.
- **Length 3** (64 possible markers): each appears in about 86% of strands, and no 3-letter marker can meet the GC rule (its GC content can only be 0, 33, 67 or 100%).

Why not 9–11, and why 12 was run only once

- At 8 bases accidental matches are already rare (0.2% of strands), so longer markers add little uniqueness, while each extra base raises overhead by about 3 points.
- Length 12 was trained once as an **upper-bound check**. It gave the best accuracy (62.3% at 8% deletion) but needed 62.1% just to break even, ending at +0.4% in stored data.
- Lengths 9–11 would fall between 8 (+1.3%) and 12 (+0.4%) in stored data, so they were not worth the extra training runs.

Next steps

- Stage 5: paired statistical tests on the same test strands and the final graphs.
- Repeat training runs (new seed) for lengths 6, 7, 8 and no markers, to confirm length 6's lead.
- Optional: end-block controls for the short markers, to check that spacing still matters at 4–6 bases.